

The Impact of Social Media on Adolescents' Health

Jarin S Wadiwalla

The Montessori Centre for Work and Study

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Introduction

Social media has become a pillar of 21st century culture, and has become central to modern communication and interaction. According to a 2024 statistical compilation by Forbes, 4.9 billion people use social media worldwide; over 62 percent of the world's population. Globally, the typical individual dedicates approximately 145 minutes each day to social media engagement. To provide context, if this usage pattern were sustained over the average lifespan of 73 years, it would accumulate to a remarkable 5.7 years of one's life! (Forbes 2023) Adolescents, in particular, find it difficult to imagine their lives without social media. A 2023 Pew Research Center survey of 1,453 American teenagers ages 13 to 17 revealed that nearly half of the teenagers self-reported using social media "almost constantly" throughout each day. The survey determined that the top 5 online platforms that teens frequent include: YouTube, TikTok, Instagram, Snapchat, and Facebook. (Anderson, 2023)

The occurrence of major depressive disorder and symptoms of depression has risen among adolescents in the United States, and there has been a significant increase in rates of suicide attempts and deaths among adolescents over the past two decades. (Burstein et al., 2019) (Hedegaard et al., 2018) (Mojtabai, Twenge et al., 2019) There are suggestions among researchers that the rise in depression could be linked to the rapid rise in social media usage. Nevertheless, there's limited evidence from nationally representative samples to support this claim, and it's unclear whether decreasing social media usage could impact the prevalence of mental health issues on a national scale. (Riehm et al., 2019) A 2019 longitudinal study by Riehm et al., of 7,594 adolescents found there to be an association between adolescents who spend more than 3 hours per day using social media and an increased risk for mental health problems, especially internalizing problems like symptoms of anxiety and depression.

While social media offers opportunities for connection, self-expression, and information sharing, it also presents significant challenges and risks that can have profound social and psychological impacts on young people. Adolescents can suffer a variety of symptoms linked to their frequent social media use. For example, heavy use of social media, and other forms of media, may result in sleep deprivation, which has a negative influence on their cognitive ability, academic performance, and socio-emotional functioning. Moreover, smartphones and social media have been associated with increased mental distress, self-harming behaviors, body image issues, and suicidality, according to evidence from numerous cross-sectional, longitudinal, and empirical studies. (Khalaf, et al., 2023) There is also an increased concern of cyber bullying, feedback-seeking habits, and a connection has been linked to depression symptoms. On the contrary, ethical social media use can increase opportunities to boost self-esteem, have more connection with peers, promote healthy habits, and access useful information. (Khalaf, et al., 2023)

This literature review will explore current research on the impacts of social media on the social and emotional health of adolescents, a brief overview of adolescent brain development research, what is being done to help mitigate the negative effects of social media for adolescents, and what society can do moving forward. Furthermore, a cross comparison of the developmental needs of 21st century adolescents, and those defined by the 18th century works of Maria Montessori, will be postulated.

Negative Impacts of Social Media Use on Adolescents' Health

Researchers have been studying the impacts of social media on the social and emotional health of adolescents for years. A scoping review of 43 papers from around the world by Azem et al., (2023) revealed a connection between social media use and depression, as well as other

negative outcomes such as anxiety, poor sleep, low self-esteem, and social and appearance anxiety.

Poor Sleep: A cross-sectional study in Norway found that among 9,846 adolescents, 90% used digital devices an hour before bedtime, which was associated with sleep onset latency and shorter sleep duration resulting in tiredness during the day. (Krishnan et al., 2020) Results from a longitudinal study conducted in Switzerland over the course of four years by Foerster et. al (2019) “found consistently elevated odds of developing various sleep problems and general health symptoms over one year for adolescents who report nocturnal mobile phone-related awakenings or high amounts of screen time.”

Social and Emotional Challenges: Additionally, social media can exacerbate issues related to cyberbullying and online harassment, which can have devastating consequences for adolescent mental health. The anonymity and distance afforded by the online environment can embolden individuals to engage in harmful behavior. Cyber victimization has been shown to correlate with higher rates of self-inflicted damage, suicidal ideation, and a variety of other internalizing and externalizing issues. (Khalaf, et al., 2023) Internalizing and externalizing issues refer to two overarching categories of behavioral problems. An internalizing problem behavior is focused on the inner person, with symptoms like anxiety, depression, or withdrawal, while an externalizing problem behavior deals with the social environment, with behaviors like hyperactivity, impulsivity, and deviance. (Nesi & Prinstein, 2015)

Symptoms of Depression: Nesi & Prinstein (2015) explain that researchers have pinpointed a gap in standing research and identified the need to look at specific online behaviors, as well as individual, offline characteristics, that may help explain associations between psychological changes and frequent technology use. Surveying a pool of 619 eighth and ninth

graders, Nesi & Prinstein examined the association of social media with symptoms of depression, while also measuring two offline characteristics of the individual: gender and popularity. The study found that the association between social comparison and feedback-seeking behaviors on social media and depressive symptoms is particularly strong among adolescents low in popularity and who identify as female.

Self-Esteem and Body Image: Another significant impact of social media on adolescents is its influence on *self-esteem and body image*. Exposure to idealized and unrealistic representations of beauty and success can contribute to feelings of inadequacy and low self-worth, particularly among young people who are still developing their sense of identity. This can manifest in behaviors such as excessive comparison, self-objectification, and seeking validation through likes and comments. This leads some young people to make unhealthy comparisons between themselves and others about their appearance, achievements, or abilities. (Pedalino & Camerini 2022) Through cross-sectional self-report of 297 Italian female adolescents and young women, Pedalino & Camerini (2022) concluded, “Our final model results show that browsing on Instagram was associated with lower levels of body appreciation, fully mediated by upward social comparison with social media influencers, not close or distant peers.”

Positive Impacts of Social Media Use on Adolescents' Health

Despite these risks, social media also presents opportunities for positive social and psychological development among adolescents. It can facilitate connections with like-minded individuals, provide access to valuable resources and support networks, and serve as a platform for self-expression and creativity. The unique features of social media ecosystems enable innovative approaches to mental health support, including mental health screening and intervention tools. (Reid et al., 2016) Khalaf et al., (2023) also found that despite concerns raised

by adults about the influence of media on youth, there were also opportunities for mental health awareness and intervention on social media platforms because of the sheer accessibility and global reach. Pedalino & Camerini (2022) found that celebrities on social media have an opportunity to positively impact the health of adolescents through body positivity campaigns. These campaigns could nurture self-esteem, cultivate emotional support, and address body image concerns of adolescents, and steer their attention away from harmful beauty standards perpetuated by social platforms. In their systematic review of several papers, Khalaf, et al. (2023) gathered that, “online friends can be a significant source of social support, but in-person social support appears to provide greater protection against persecution.”

The Impressionable Adolescent Brain

The adolescent brain is in a uniquely vulnerable and crucial stage. Modern neuroscience tells us that the brain goes through important structural maturation during adolescence. This may explain why adolescents are a particularly vulnerable population susceptible to the possible risks of frequent social media use.

The gray matter in the brain matures from back to front, meaning that brain areas such as the prefrontal cortex— in charge of higher cognitive functions such as behavioral control, planning, and assessing the risk of decisions— mature later than areas associated with sensory and motor tasks. (Konrad et al., 2013) Recent studies have also provided evidence that adolescents have heightened responsiveness to incentives and socioemotional context during a time when impulse control is relatively immature. (Berryman et al., 2018) The teenage brain can easily adapt to its surroundings, which means that what happens in the environment can have a big impact on how the brain develops. This is important because it allows teenagers to learn and

grow both intellectually and emotionally. However, it also means that negative influences from the environment can greatly affect them. (Konrad et al., 2013)

Autopsy findings suggest that the changes observed in gray matter during adolescence are attributed to synaptic pruning. During childhood, numerous synapses are formed, but many are eliminated during adolescence, a process known as synaptic pruning. This pruning occurs in response to experience, meaning that the synapses that remain are those that are frequently utilized. Additionally, imaging studies indicate that adolescents exhibit heightened activity in limbic areas during emotional situations. For instance, Galvan et al., (2006), discovered that “adolescents show increased activation in the nucleus accumbens when anticipating a reward compared to children and adults.” (Konrad et al., 2013, para. 16)

In summary, the adolescent brain undergoes significant development, marked by an early maturation of the limbic and reward systems compared to the still-developing prefrontal control system. This imbalance could explain the typical emotionally reactive behavior seen in adolescence and might contribute to risky behaviors. The brain's time of reorganization also makes it highly responsive to environmental influences, positive and negative. (Konrad et al., 2013)

Maria Montessori on the Optimal Developmental Environment of the Adolescent

As Montessori practitioners, a brief reflection on Maria Montessori’s early work in relation to this modern digital conundrum and the developmental needs of the adolescent is necessary. In the mid 1900s, Maria Montessori gave lectures on the reform of education for the period of adolescence. She describes the environment that is needed to best nurture the social and emotional development of the adolescent.

“...during the difficult time of adolescence it is helpful to leave the accustomed environment of the family in the town and go to quiet surroundings in the country, close to nature. Here, an open-air life, individual care, and a non-toxic diet, must be the first considerations in organizing a “centre for study and work.” (Montessori, *From Childhood to Adolescence*, p. 63)

She also says that the most important thing, above all else, is that the adolescent should have a life of “activity and variety.” (Montessori, *From Childhood to Adolescence*, p. 67) Although smartphones and social media did not exist at the time of Montessori’s writings, it is clear that she advocated for an adolescent environment surrounded by nature, plenty of physical movement, and interactions with their peers, which would exclude the sedentary nature of using personal electronic devices indoors and in solitude. Adolescents’ reporting that they are “almost constantly” online is certainly not living a life of activity and variety. Montessori believed that during adolescence, individuals undergo significant emotional and social transformations, and it is crucial to create environments that respect and nurture their emerging identities and interpersonal relationships.

“The chief symptom of adolescence is a state of expectation, a tendency towards creative work and a need for the strengthening of self-confidence.” (Montessori, *From Childhood to Adolescence*, p. 63)

Twenty-first century research is in line with Montessori’s beliefs about the developmental tasks of the adolescent. Crone & Dahl (2012) described them as “forming close relationships, discovering their identity, envisioning their future, gaining independence, building self-esteem, learning self-discipline, and mastering social interaction skills.”

Literature Review Conclusion

Ultimately, the relationship between social media use and adolescent mental health remains a very complex issue and requires further study to provide causation of negative or positive impacts. While studies indicate potential risks and benefits, ongoing research is necessary to understand the long-term effects of social media on adolescent well-being. Digital media's rapid evolution necessitates continual reassessment of its impact on teenagers' mental health, particularly in light of emerging concerns and opportunities. As social media continues to shape young people's experiences, comprehensive research is essential to inform effective strategies for promoting mental health and addressing potential challenges in this dynamic landscape. (Nesi & Prinstein, 2015) There is currently a lack of empirical research on how adolescents comprehend social media as a body of wisdom. (Khalaf, et al., 2023) Existing studies are mostly cross-sectional (from a single point in time) and use self-report for adolescent media intake. Cross-sectional studies make it difficult to infer causation because they are observational in nature and are known as descriptive research, not causal or relational. More longitudinal studies are needed to collect data on the same group of participants over time to observe any changes.

The scoping review from Azem et al., (2023) suggests that there is an association between social media use and depression among adolescents but a systematic review needs to be conducted to understand the associations further. Nesi & Prinstein (2015) noted that a number of studies have found no association between frequency of technology use and general well being or depressive symptoms. This highlights the inconclusive nature of attempts to characterize overall associations between technology use and psychological outcomes. Nesi & Prinstein (2015) suggest that it may not be the quantity of social networking site usage that over time

influences depressive symptoms, but perhaps the positive or negative quality of peer interactions that occur on these platforms. For this reason, researchers have identified the need to examine offline characteristics alongside adolescents' social media habits to help explain associations between any psychological changes. However, there is a longitudinal study that shows a causal link between the use of Facebook and negative impact on mental health from Luca Braghieri and Alexey Makarin out of Cambridge, Massachusetts. The results from their research are not discussed in this review because the age group of participants were undergraduate college students, not adolescents.

Current Societal Responses to the Social Media Mental Health Crisis

Beyond the research, real families are living the consequences of social media's impact on young people, and a growing wave of legal, legislative, and community efforts is rising to meet the crisis. Unfortunately, many families across the U.S. alone have suffered horrendous tragedies which have been directly linked to information that their children interacted with on various social media apps. At the forefront of cases like these is the *Social Media Victims Law Center*, a U.S. law firm that secured meaningful victories for some families in 2026, even as countless others continue their long pursuit of justice.

In addition, over the last several years, the U.S. Senate and EU Parliament have held hearings about social media and mental health. Mark Zuckerberg, CEO of Meta, opened his speech during the U.S. senate hearing with this statement, "The existing body of scientific work has not shown a causal link between using social media and young people having worse mental health." Here you see Zuckerberg taking advantage of the difficult nature of these studies. According to a recent article by Reuters, 42 U.S. authorities filed lawsuits against Meta, the owners of Facebook and Instagram, over the negative impacts of social media on children's

health. Eight states and Washington, D.C. are filing similar lawsuits against the California-based company. Other social media companies face hundreds of lawsuits in the United States filed by families, young people, and school districts over the impact on mental health, but this is the biggest action to date. (Reuters, 2023)

In May 2023, the United States Surgeon General, Dr. Vivek Murthy, released an advisory about the effects of social media use on youth mental health. The advisory calls for policymakers, technology companies, researchers, families, and young people to understand the benefits and harms of social media platforms and help create safer, healthier online environments for children. The advisory contains 14 pages of research and information, and ends with a call to action page for each group mentioned above. This step increased awareness and suggested actions to help mitigate the potential harms of social media for youth.

There are local grassroots organizations in the U.S. like *Wait Until 8th*, which empower parents to band together to delay giving their children smartphones and social media until 14-years-old, or “8th” grade. The pledge is advocated in local school communities where parents can sign-up together. It works best when large portions of a school community commit together. According to their website, more than 155,000 U.S. families have said yes to waiting on smartphones and social media for their children. If students need cellphones for communication before eighth grade, *Wait Until 8th* suggests using basic cellphones with calling capability only. The pledge can be implemented internationally as well.

When families are ready to introduce smartphones and social media, it can be done in a slow and meticulous way. Families can work with their healthcare providers to reduce the hazards by using open, nonjudgmental, developmentally appropriate tactics, including education and practical problem solving. (Khalaf, et al., 2023) Establishing family media use plans can

support open family communication, create consistent media rules, outline content boundaries, promote digital literacy, and balance online time with other fruitful activities. (Reid et al., 2016)

In 2024, prominent U.S. social psychologist Jonathan Haidt published *The Anxious Generation: How the Great Rewiring of Childhood Is Causing an Epidemic of Mental Illness*, a book examining the role of smartphones and social media in the dramatic rise of youth mental illness. He goes into this very subject and suggests four simple rules for all families to adopt. He also suggests that the most successful way to create change around device usage and adolescents is socially through group agreements, like what the organization *Wait Until 8th* is doing.

While causal evidence is lacking in current studies focusing on adolescents, the prominence and immediacy of this issue cannot be ignored. Adolescents need support to form healthy online habits to minimize potential harms. Efforts in research and advocacy should persist to raise awareness about these challenges and safeguard the futures of youth globally.

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